

SentinelX

AI-Assisted Organizational Threat Triage Platform

SentinelX is a responsible AI-assisted organizational threat triage platform designed for NGOs, intervention organizations, investigators, and institutional safety teams supporting vulnerable individuals facing:

- stalking,
- coercion,
- cyber-harassment,
- exploitation,
- grooming,
- sextortion,
- and contextual abuse.

Rather than replacing human judgment, SentinelX was intentionally designed around **human-in-the-loop AI governance**.

The platform combines:

- contextual NLP,
- on-device threat classification,
- Explainable AI synthesis,
- and analyst-supervised intervention workflows

to help organizations transform fragmented safety signals into structured, reviewable intervention intelligence.

Key Features

1. SOS Emergency Escalation

A non-AI emergency flow where users can trigger urgent alerts through a 3-second long press.

Alerts are routed directly to:

- emergency contacts,
 - NGO dashboards,
 - and intervention review queues.
-

2. Detect Threat (On-Device NLP)

A contextual spoken-threat detection pipeline using:

- speech transcription,
- on-device Bi-LSTM classification,
- and contextual NLP threat detection.

Important Privacy Design

Audio is:

- never stored,
- never uploaded,
- and discarded immediately after transcription.

Only the classification result is transmitted.

3. Safe Chat + Evidence Upload

A guided 4-step contextual reporting system that allows users to:

- describe incidents,
- categorize abuse scenarios,
- indicate escalation patterns,
- and upload evidence files.

Reports remain inactive until an NGO analyst explicitly requests AI analysis.

4. Human-in-the-Loop AI Workflow

SentinelX intentionally prevents autonomous intervention.

AI:

- does NOT contact authorities,
- does NOT escalate cases,
- does NOT make enforcement decisions.

Human analysts remain responsible for:

- interpretation,
- escalation,
- prioritization,
- and intervention decisions.

Technology Stack

Mobile Application

- React Native Expo

NGO Dashboard

- React

Backend

- Flask
- Python

AI / NLP

- TensorFlow
- Keras
- Bi-LSTM
- Gemini API

Database

- MongoDB

Hosting

- Railway
-

Repository Structure

```
SentinelX
|
├── Guide.pdf
|
├── Pitch_Deck/
|   ├── SentinelX_Pitch_Deck.pdf
|   └── SentinelX_Architecture_Diagram.png
|
└── Demo/
```

```
| SentinelX_Demo_Video.mp4
| Demo_Script.pdf
| Demo_Screenshots/
|   | mobile_safechat.png
|   | analyst_dashboard.png
|   | analyze_report_flow.png
|   | xai_panel.png
|   | threat_alert_queue.png
|
| Documentation/
|   | About_The_Project.pdf
|   | AI_Architecture_Explanation.pdf
|   | Responsible_AI_Guardrails.pdf
|   | Human_in_the_Loop_Design.pdf
|   | Technical_Stack.pdf
|   | Data_Sources.pdf
|
| Prototype/
|   | Android_APK/
|   |   | SentinelX.apk
|   |
|   | Dashboard_Screens/
|   |   | dashboard_home.png
|   |   | pending_analysis.png
|   |   | xai_report_panel.png
|   |   | triaged_case.png
|   |
|   | Mobile_App_Screens/
|   |   | safechat_step1.png
|   |   | safechat_step2.png
|   |   | safechat_step3.png
|   |   | safechat_step4.png
|   |   | sos_screen.png
|   |   | detect_threat.png
|
| Source_Summary/
|   | Folder_Structure.png
|   | Key_Code_Explanation.pdf
|   | API_Workflow_Explanation.pdf
|
| Team/
|   | Team_Photo.jpg
|   | Team_Members.pdf
|   | Team_Roles.pdf
```

Important Files for Reviewers

Start Here

Reviewers should begin with:

Guide.pdf

This document contains:

- complete prototype navigation instructions,
 - reviewer walkthrough,
 - dashboard testing steps,
 - and AI workflow explanation.
-

Pitch Materials

Located in:

Pitch_Deck/

Contains:

- final presentation deck,
 - architecture diagram,
 - and system workflow visuals.
-

Demo Files

Located in:

Demo/

Contains:

- demo video,
- demo narration script,
- and supporting screenshots.

Documentation Files

Located in:

Documentation/

Contains:

- About the Project,
- AI Architecture Explanation,
- Responsible AI Guardrails,
- Human-in-the-Loop Design,
- Technical Stack,
- and Data Sources.

Prototype Files

Located in:

Prototype/

Contains:

- Android APK,
- mobile UI screenshots,
- and dashboard screenshots.

Source Summary

Located in:

Source_Summary/

Contains:

- codebase overview,
- API workflow explanations,

- and architecture breakdowns.
-

Team Information

Located in:

Team/

Contains:

- team photograph,
 - team member information,
 - and project role breakdowns.
-

Usage Examples

Example 1 — SOS Flow

Purpose

Immediate emergency escalation.

How to Use

1. Open mobile app
 2. Press and hold SOS button for 3 seconds
 3. Alert routes instantly to:
 4. NGO Dashboard
 5. emergency contacts
-

Example 2 — Detect Threat

Purpose

On-device spoken threat classification.

How to Use

1. Toggle:

Detect Threat = ON

1. The app:
2. transcribes speech locally,
3. classifies contextual threats,
4. discards audio immediately.
5. Threat signals appear on NGO dashboard.

Example 3 — Safe Chat

Purpose

Structured contextual reporting.

How to Use

Users complete:

1. Category Selection
2. Narrative Description
3. Timing Tag
4. Frequency Tag

Optional:

- image upload,
- video upload,
- audio upload.

Submitted reports appear as:

Pending Analysis

until reviewed by an analyst.

Prototype Navigation Guide

This section provides a step-by-step walkthrough for reviewers testing the SentinelX prototype.

Mobile App SIGN UP

Step 1

Download the APK located in:

Prototype/Android_APK/

Install on Android device.

Step 2

Open the mobile application.

Step 3

Allow SentinelX notification permissions (optional).

Step 4

Click:

Continue with Phone Number

Step 5

Enter phone number and click:

Send Code

Step 6

For this MVP prototype, use hardcoded OTP:

2026

Step 7

Allow location access.

Step 8

Fill in:

- First Name
 - Last Name
 - Email
-

Step 9

Add at least one emergency contact.

Important Note

If the app gets stuck during onboarding:

- restart the app,
 - and complete setup more quickly.
-

NGO Dashboard Login

Dashboard URL:

<https://besafe-server-production.up.railway.app>

Login or sign up using NGO analyst credentials.

Important Dashboard Note

Ensure your current location is enabled.

After Login Reviewers Will See

- Overview Page
 - Alert Queue
 - Safe Chat Reports
 - AI Analysis Controls
-

Mobile App Walkthrough

The mobile application contains three major flows:

1. SOS
 2. Detect Threat
 3. Safe Chat + Evidence Upload
-

A. SOS Flow (No AI)

Purpose

Immediate emergency escalation without AI involvement.

Steps

Step 1

Press and hold the SOS button for approximately 3 seconds.

Step 2

The system sends:

- emergency alerts,
- dashboard notifications,
- contextual emergency information.

Step 3

Open the NGO dashboard and observe:

- incoming SOS alert,
 - timestamp,
 - priority indicator,
 - and user information.
-

Important Design Note

The SOS flow intentionally uses:

NO AI

This preserves direct human-controlled escalation.

B. Detect Threat Flow (On-Device AI)

Purpose

Contextual spoken threat classification using NLP.

Steps

Step 1

Return to Home Screen.

Step 2

Toggle:

Detect Threat

to ON.

Step 3

The app begins:

- local audio capture,
- local transcription,
- on-device Bi-LSTM classification.

Step 4

If a threat is detected:

- a threat signal is routed to dashboard.

Step 5

Open dashboard and observe:

- threat alert entry,
 - transcript snippet,
 - classification result.
-

Important Privacy Design

Audio is:

- never stored,
- never uploaded,
- and discarded immediately after transcription.

Only the classification result leaves the device.

C. Safe Chat + Evidence Flow

Purpose

Structured contextual abuse reporting.

Steps

Step 1

Open:

Safe Chat

from Home Screen.

Step 2 — Category Selection

Select:

- stalking,
- cyber-harassment,
- coercion,
- exploitation,
- physical tracking.

Step 3 — Narrative Description

Enter contextual description.

Step 4 — Timing

Select:

- today,
- this week,
- months ago.

Step 5 — Frequency

Select:

- first time,
- a few times,
- repeated escalation.

Step 6 — Evidence Upload

Optional uploads:

- image,
- video,
- audio.

Step 7 — Submit Report

Report appears on dashboard as:

Pending Analysis

NGO Dashboard Walkthrough

The dashboard serves as the primary analyst workspace.

It contains:

- real-time alerts,
- Safe Chat reports,
- Explainable AI panels,
- AI analysis controls,
- and analyst workflows.

A. Viewing Incoming Alerts

Navigate to:

Alerts

Analysts can review:

- SOS alerts,
- Detect Threat signals,
- timestamps,
- contextual information,
- priority indicators.

Clicking an alert opens:

Detail Panel

which displays:

- transcript snippets,
- timestamps,
- contextual details,

- analyst controls.
-

B. Reviewing Safe Chat Reports

Navigate to:

Reports

Reports initially appear as:

Pending Analysis

At this stage:

- no AI analysis has occurred,
- no synthesis has been generated.

This is intentional.

C. Human-Triggered AI Analysis

Important Responsible AI Workflow

AI analysis only begins after explicit analyst authorization.

Steps

Step 1

Open Safe Chat report.

Step 2

Click:

[Analyze Report]

Step 3

The backend:

- retrieves report,
- processes metadata,
- sends contextual payload,
- receives explainable synthesis.

Step 4

Status updates from:

Pending Analysis

to:

Triaged

Step 5

Explainable AI panel becomes visible.

The panel displays:

- contextual synthesis,
- escalation observations,
- explainable reasoning,
- risk indicators.

D. Human-in-the-Loop Workflow

SentinelX intentionally prevents autonomous intervention.

Reviewers should note:

- AI never contacts authorities,
- AI never escalates automatically,
- AI never performs enforcement actions.

Human analysts remain responsible for:

- interpreting outputs,

- validating contextual findings,
 - making intervention decisions.
-

Recommended Reviewer Flow

For best demonstration experience:

1. Mobile App Home Screen
2. SOS Flow
3. Detect Threat Flow
4. Safe Chat Submission
5. NGO Dashboard Alert Queue
6. Safe Chat Report Queue
7. Analyze Report Trigger
8. Explainable AI Panel
9. Analyst Action Controls

This demonstrates:

- multi-signal ingestion,
 - contextual NLP,
 - Explainable AI,
 - human-supervised AI governance.
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Responsible AI Design

SentinelX was intentionally designed to reduce:

- automation bias,
 - privacy violations,
 - over-reliance on AI,
 - autonomous escalation risks.
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Core Safeguards

- Human-triggered AI only
- No autonomous intervention
- Explainable AI synthesis
- On-device transcription
- Audio deletion after transcription
- Transparent analyst review workflow

Contributing Guidelines

We welcome contributions from:

- developers,
- researchers,
- NGOs,
- responsible AI practitioners,
- and safety-focused organizations.

Contribution Areas

- NLP improvements
- UI/UX enhancements
- Dashboard improvements
- Security hardening
- Explainable AI tooling
- Accessibility support

How to Contribute

Step 1

Fork repository.

Step 2

Create feature branch:

```
git checkout -b feature-name
```

Step 3

Commit changes:

```
git commit -m "Added new feature"
```

Step 4

Push branch:

```
git push origin feature-name
```

Step 5

Open Pull Request.

AI Tools Used

AI Models

- Bi-LSTM
- Gemini API

Frameworks

- TensorFlow
- Keras

AI-Assisted Development

- ChatGPT
 - GitHub Copilot
 - Claude
 - Gemini
 - Lovable
-

Dataset Information

The NLP model was trained using:

- public NLP datasets,
- contextual threat examples,
- synthetic escalation narratives.

Dataset size:

100,000+ labeled entries

Dataset split:

70% Training
20% Validation
10% Testing

Team

Fredrick

Team Lead / System Design / AI Research

Tomna

Machine Learning Engineer

Solomon

Mobile Application Developer

Habeeb

UI/UX Designer

Adebayo

Asst AI/ML lead

Important Research Note

SentinelX is currently an MVP / pilot-stage prototype created during the USAII Hackathon.

The system is intended for:

- research,
- prototyping,
- organizational testing,
- and responsible AI experimentation.

It should not replace:

- law enforcement,
 - emergency response systems,
 - or professional crisis intervention services.
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License

This repository is currently released for:

- educational,
- research,
- and hackathon demonstration purposes only.